

Chemistry V2 — Core Study Guide

CHEMISTRY V2 | 1

Everything for this chapter in one place. The teaching comes first, then practice questions, then quick answers you can check yourself, then the full worked answers, and last a one-page sheet you can print and keep.

Name the role only after you have tracked the change

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.

Treat the equation as a before/after map of the same chemical entities.

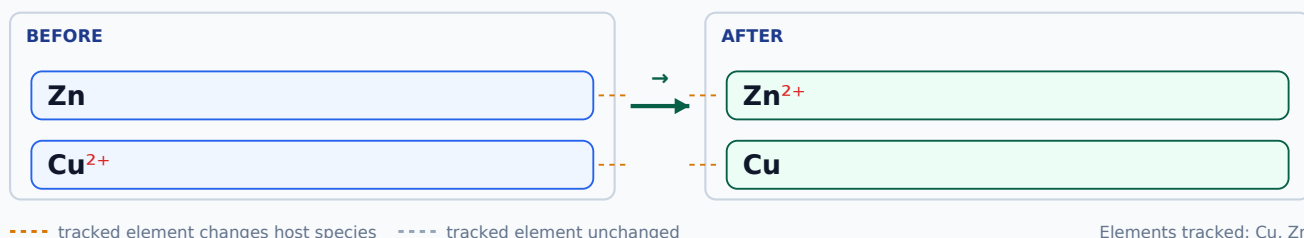
The idea in plain words

A role is proved from what a species does in this equation, not from reagent reputation.

Picture it like this

- Track changing species first.
- Separate unchanged or spectator information.
- Attach a role only after the change is established.
- Use the symbolic representation.
- Read the formula notation explicitly.
- Read the ionic charge explicitly.

Track each element from before to after



Sort the species before naming any role



Reactant species	Zn Cu²⁺
Species that changes	fill only from the evidence above
Spectator (if present)	fill only from the evidence above
Product species	Zn²⁺ Cu

Attach a role only after identifying the changing reactant species.

The rule you will use

Equation/source evidence is authoritative; memorized reagent reputation is not.

Why it works

- Identify reactant species.
- Track each through the equation.
- Mark change/no change.
- Attach the role to the reactant evidence.
- Verify spectators are not mislabelled.

Watch one worked out

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Work on this practice instance before attempting the transfer

questions.

- Track the reactant species first.
- Use its recorded change/effect as evidence.
- Attach the requested role and exclude spectators.

Does the answer make sense?

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

If you get stuck

List the starting substances down the left and the finishing ones down the right, then join each substance to itself with a line before you name any role.

Common mistake to avoid

An easy way to get this wrong: A familiar reagent has one permanent role.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: A minimal pair differing on exactly one decisive feature.

- Return to species tracking.
- Name the self-change before the role.
- Retry with a new equation/representation.

Try that one again: After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A

Zn

Predict this case before reading on.

CASE B

Zn

Predict this case before reading on.

Try it with help

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Use the first-move and picture cues shown here.

Try it with less help

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. One cue has been removed; choose the next move yourself.

Now try it on your own

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Try it independently using only the information in the task.

Check your answer

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- ☐ Check that the overall charge is preserved or interpreted correctly.
- ☐ Check that the same chemical species is being tracked.
- ☐ Check that the final conclusion follows from the evidence and rule used.

Where this shows up next

You will meet this again in: same role logic in a new equation. Attempt the transfer questions only after this learning step.

Check that the atoms balance

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.

Frame conservation as an accounting check on a represented process.

The idea in plain words

Count the exact thing the route says must be conserved; different ledgers answer different questions.

Picture it like this

- Create before/after ledgers for the required invariant.
- Keep atom count, charge and species identity as separate ledgers.
- Use the symbolic representation.
- Read the formula notation explicitly.

Atom ledger before and after



Element	Before	After	Matched?
H	4 	4 	✓ same
O	2 	2 	✓ same

Fill or compare the required ledger and locate the first mismatch.

The rule you will use

Choose the conservation check the question actually needs.

Why it works

- Name the ledger.
- Count before/after.
- Compare.
- Locate the first mismatch.
- Recheck after repair.

Watch one worked out

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Work on this practice instance before attempting the transfer questions.

- Choose the atom-count ledger.
- Count each required atom before and after.
- Compare counts and identify any mismatch.

Does the answer make sense?

- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

If you get stuck

Draw two columns, give each element its own row, and count that element on the left and on the right before you compare anything.

Common mistake to avoid

An easy way to get this wrong: A balanced-looking expression has passed every chemical check.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: Contrast an atom-count mismatch with a charge or species-identity issue.

- Separate the ledgers.
- Repair the failed ledger only.
- Run the remaining required checks.

Try that one again: Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A



Predict this case before reading on.

CASE B



Predict this case before reading on.

Try it with help

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Use the first-move and picture cues shown here.

Try it with less help

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. One cue has been removed; choose the next move yourself.

Now try it on your own

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Try it independently using only the information in the task.

Check your answer

- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

☐ Count each required atom on both sides.

☐ Check that the final conclusion follows from the evidence and rule used.

Where this shows up next

You will meet this again in: new process with the same conservation obligation. Attempt the transfer questions only after this learning step.

Check the rule, the condition and the exception

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.
Show the default rule and the gate that can block or modify it.

The idea in plain words

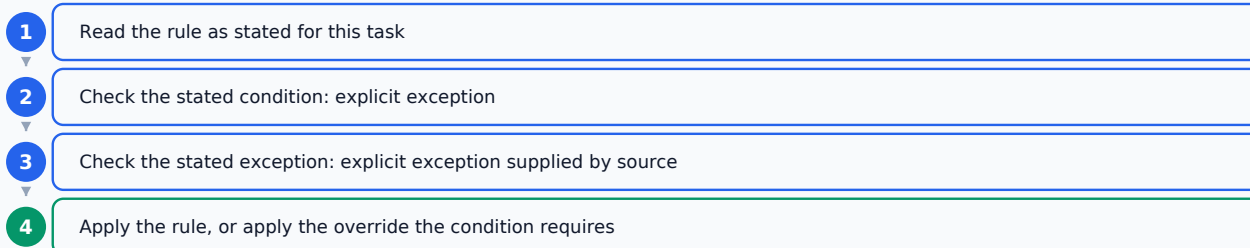
A remembered rule is incomplete until the stated condition or exception is checked.

Picture it like this

- Rule cue first.
- Condition/exception gate second.
- Decision only after the gate check.
- Use the symbolic representation.

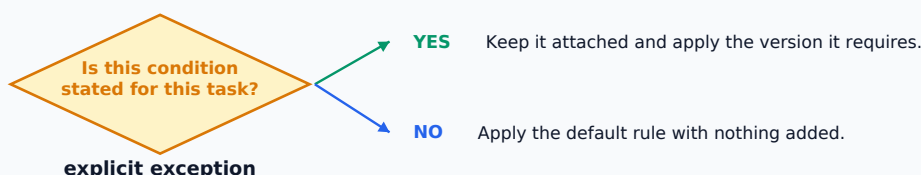
Order in which the rule is consulted

Rule selection before condition/exception decision.



Stop at the first step whose stated condition decides the case.

Condition gate before the rule applies



The rule you will use

Only upstream-recorded rules, conditions and exceptions may populate the gate.

Why it works

- Select the rule.
- Read the condition/exception.
- Test applicability.
- Apply or withhold the rule.
- Verify the condition remains attached.

Watch one worked out

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Work on this practice instance before attempting the transfer questions.

- Select the recorded rule.

- Check the condition/exception gate.
- Apply or withhold the rule accordingly.

Does the answer make sense?

- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

If you get stuck

Underline the condition written with the reaction, then ask whether the usual rule still holds while that condition is there.

Common mistake to avoid

An easy way to get this wrong: Remembering the default rule is enough.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: A minimal pair differing on exactly one decisive feature.

- Re-run the gate.
- Name the decisive condition.
- Retry on a close contrast.

Try that one again: Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A

Predict this case before reading on.

CASE B

Predict this case before reading on.

Try it with help

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Use the first-move and picture cues shown here.

Try it with less help

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. One cue has been removed; choose the next move yourself.

Now try it on your own

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Try it independently using only the information in the task.

Check your answer

- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- ☐ Check that every stated condition or exception is still attached.
- ☐ Check that the final conclusion follows from the evidence and rule used.

Where this shows up next

You will meet this again in: same rule with a less obvious condition/exception. Attempt the transfer questions only after this learning step.

Decide what kind of change happened, from the evidence

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.

Anchor classification in the supplied evidence or process description.

The idea in plain words

Name the evidence before naming the class.

Picture it like this

- Evidence features lead to a criterion; the criterion leads to the class.
- Use the stated macroscopic evidence.
- Use the symbolic representation.
- Keep the recorded observation separate from interpretation.
- Keep physical-state information attached.

What was actually observed

i RECORDED AT THE MACROSCOPIC LEVEL

Expose the source-recorded observable evidence before a chemical claim or classification.

Nothing has been inferred yet — this box stays empty until you decide.

Evidence, link, claim

EVIDENCE

Separate observation, claim and reasoning link so overclaiming is inspectable.

LINK

Which evidence supports which part of the claim.

CLAIM

Read observation first, then test each claim against the evidence link.

An inference is only as strong as the recorded evidence behind it.

The rule you will use

Classify only from the stated evidence and the rule taught here.

Why it works

- Extract evidence.
- Select the criterion.
- Compare.
- Classify.
- Verify that the evidence supports the label.

Watch one worked out

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Work on this practice instance before attempting the transfer questions.

- Extract the supplied evidence.
- Apply the authorized criterion.
- Classify only after the evidence matches the criterion.

Does the answer make sense?

- Check that the final conclusion follows from the evidence and rule used.

If you get stuck

Write down the single observation that decides this, then test it against the rule before you write any label.

Common mistake to avoid

An easy way to get this wrong: A familiar word in the prompt is enough to classify the process. The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: Use two cases with similar wording but different decisive evidence.

- Return to the evidence.
- Apply the criterion explicitly.
- Retry on the near contrast.

Try that one again: State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A

Predict this case before reading on.

CASE B

Predict this case before reading on.

Try it with help

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Use the first-move and picture cues shown here.

Try it with less help

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. One cue has been removed; choose the next move yourself.

Now try it on your own

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Try it independently using only the information in the task.

Check your answer

- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

☐ Check that the final conclusion follows from the evidence and rule used.

Where this shows up next

You will meet this again in: near-neighbour classification with new evidence. Attempt the transfer questions only after this learning step.

Tell an ion charge apart from a subscript

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.

Use a side-by-side notation pair rather than a warning alone.

The idea in plain words

The position of the numeral is evidence for what it means.

Picture it like this

- Hold element symbols fixed.
- Change one numeral position only.
- Use the symbolic representation.
- Read the formula notation explicitly.
- Read the ionic charge explicitly.

Anatomy of the formula

SUBSCRIPT 4

atoms inside one species



IONIC CHARGE $^{2-}$

net charge on the whole ion

Notation position and its local chemical meaning.

Charge position versus subscript position

Same species, same symbols — only the position you read changes.

CASE A



The lowered numeral counts atoms inside the species.

CASE B



The raised numeral with a sign is the net charge on the whole ion.

Compare two cases and state what the numeral means in each position.

The rule you will use

Read coefficient, subscript and charge as different notation roles.

Why it works

- Present a minimal pair.
- Identify the only changed feature.
- Explain the changed meaning.
- Retry without labels.

Watch one worked out

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Work on this practice instance before attempting the transfer questions.

- Locate the superscript charge separately from subscripts/coefficients.
- Read the ionic charge.
- Verify the species identity is unchanged.

Does the answer make sense?

- Check that the overall charge is preserved or interpreted correctly.

- Check that the same chemical species is being tracked.

If you get stuck

Circle the number you are unsure about and say where it sits: in front of the formula, low down after a symbol, or raised up after it.

Common mistake to avoid

An easy way to get this wrong: The numeral itself carries a fixed meaning wherever it appears. The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: A minimal pair differing on exactly one decisive feature.

- Circle the position.
- Name the feature.
- Restate the chemical meaning.

Try that one again: Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A



Predict this case before reading on.

CASE B



Predict this case before reading on.

Try it with help

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Use the first-move and picture cues shown here.

Try it with less help

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. One cue has been removed; choose the next move yourself.

Now try it on your own

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Try it independently using only the information in the task.

Check your answer

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

Run these checks before accepting

☐ Check that the overall charge is preserved or interpreted correctly.

☐ Check that the same chemical species is being tracked.

Where this shows up next

You will meet this again in: notation-position discrimination under reduced cues. Attempt the

transfer questions only after this learning step.

Keep the reaction conditions attached

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.
Show the default rule and the gate that can block or modify it.

The idea in plain words

A remembered rule is incomplete until the stated condition or exception is checked.

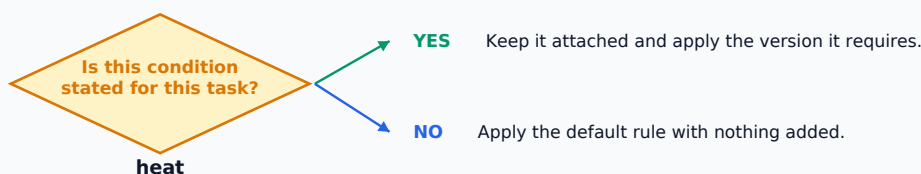
Picture it like this

- Rule cue first.
- Condition/exception gate second.
- Decision only after the gate check.
- Use the symbolic representation.
- Use the condition representation.
- Read the formula notation explicitly.

Condition gate before the rule applies



The recorded condition stays attached to the arrow.



The rule you will use

Only upstream-recorded rules, conditions and exceptions may populate the gate.

Why it works

- Select the rule.
- Read the condition/exception.
- Test applicability.
- Apply or withhold the rule.
- Verify the condition remains attached.

Watch one worked out

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Work on this practice instance before attempting the transfer questions.

- Locate the recorded condition.
- Keep it attached to the process representation.
- Check that the rewritten/read representation preserves it.

Does the answer make sense?

- Check that every stated condition or exception is still attached.

If you get stuck

Underline the condition written with the reaction, then ask whether the usual rule still holds while that condition is there.

Common mistake to avoid

An easy way to get this wrong: Remembering the default rule is enough.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: Two cases use the same rule but differ only in the decisive condition/exception.

- Re-run the gate.
- Name the decisive condition.
- Retry on a close contrast.

Try that one again: Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A

A

Predict this case before reading on.

CASE B

A

Predict this case before reading on.

Try it with help

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Use the first-move and picture cues shown here.

Try it with less help

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. One cue has been removed; choose the next move yourself.

Now try it on your own

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Try it independently using only the information in the task.

Check your answer

- Check that every stated condition or exception is still attached.

Run these checks before accepting

☐ Check that every stated condition or exception is still attached.

Where this shows up next

You will meet this again in: same rule with a less obvious condition/exception. Attempt the transfer questions only after this learning step.

Connect the apparatus, the property and the method

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.

Start from the property/evidence needed, then connect it to the apparatus or method feature.

The idea in plain words

Choose the method because of the property it exploits, not because the diagram looks familiar.

Picture it like this

- Target property → apparatus feature → expected observation/use.
- Use the stated macroscopic evidence.
- Read the figure as part of the evidence.

Property, apparatus, intended observation

PROPERTY USED

Property → apparatus feature → intended observation or use.



APPARATUS / METHOD

Connect apparatus or method choice to the source-authorized property or observation it uses.



INTENDED OBSERVATION

Trace the method choice back to the property/evidence it is meant to exploit.

The rule you will use

Only apparatus/method properties authorized by source may be used.

Why it works

- Identify target property.
- Identify apparatus feature.
- Link feature to property.
- Verify against the goal/observation.

Watch one worked out

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Work on this practice instance before attempting the transfer questions.

- Identify the target property or observation.
- Identify the apparatus/method feature.
- Explain the property-feature link.

Does the answer make sense?

- Check that the final conclusion follows from the evidence and rule used.

If you get stuck

Write down which property is being used, then decide which part of the apparatus uses it.

Common mistake to avoid

An easy way to get this wrong: Recognizing the apparatus shape is enough.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: Compare two visually similar method representations that differ on the decisive property link.

- Name the property first.

- Trace it to the apparatus feature.
- Retry with labels removed.

Try that one again: Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A

Predict this case before reading on.

CASE B

Predict this case before reading on.

Try it with help

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Use the first-move and picture cues shown here.

Try it with less help

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. One cue has been removed; choose the next move yourself.

Now try it on your own

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Try it independently using only the information in the task.

Check your answer

- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

☐ Check that the final conclusion follows from the evidence and rule used.

Where this shows up next

You will meet this again in: same property-method link in a different representation. Attempt the transfer questions only after this learning step.

Read a chemical formula before you calculate

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.

Treat the formula as a structured message, not a string of characters.

The idea in plain words

Read position first, then meaning: a lower numeral, a leading numeral and an upper charge do different jobs.

Picture it like this

- Point separately to symbol, subscript, coefficient and charge positions.
- Map each position to its local meaning.
- Use the symbolic representation.
- Read the formula notation explicitly.
- Read the ionic charge explicitly.

The rule you will use

Do not infer chemical facts beyond the notation and source-authorized rules.

Why it works

- Locate notation features.
- Name each feature.
- State its local meaning.
- Check the whole species identity.

Watch one worked out

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Work on this practice instance before attempting the transfer questions.

- Locate each notation feature by position.
- State the local meaning of each feature.
- Check the complete species notation.

Does the answer make sense?

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

If you get stuck

Copy the formula out with space around each part, then circle the number in front, each low number and any raised charge before you try to read what it means.

Common mistake to avoid

An easy way to get this wrong: All nearby numerals mean the same thing.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: A minimal pair differing on exactly one decisive feature.

- Re-anchor to position.
- Name the notation feature.
- Retry on a new source-authorized formula.

Try that one again: Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A

Predict this case before reading on.

CASE B

Predict this case before reading on.

Try it with help

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Use the first-move and picture cues shown here.

Try it with less help

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. One cue has been removed; choose the next move yourself.

Now try it on your own

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Try it independently using only the information in the task.

Check your answer

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

Run these checks before accepting

☐ Check that the overall charge is preserved or interpreted correctly.

☐ Check that the same chemical species is being tracked.

Where this shows up next

You will meet this again in: formula/charge discrimination in a new species. Attempt the transfer questions only after this learning step.

Keep what you saw apart from what it means

Get ready

Identify the target and state the first chemically meaningful move: Identify the target. Start with what was actually observed before naming a chemical conclusion.

The idea in plain words

An observation is evidence; a claim must not say more than that evidence supports.

Picture it like this

- Record observation.
- State claim separately.
- Use an explicit evidence-to-claim link.
- Use the stated macroscopic evidence.
- Keep the recorded observation separate from interpretation.

What was actually observed

i RECORDED AT THE MACROSCOPIC LEVEL

Expose the source-recorded observable evidence before a chemical claim or classification.

Nothing has been inferred yet — this box stays empty until you decide.

Observation kept apart from inference

Literal evidence versus interpreted chemical claim.

OBSERVATION — what was recorded	INFERENCE — what is claimed from it
Hold observation and inference in separate cells so one cannot be silently substituted for the other.	

Place each statement in the correct column and justify the inference link.

The rule you will use

Source observations and authorized conclusions constrain the reasoning chain.

Why it works

- Record the observation literally.
- State a candidate claim.
- Explain the link.
- Test for overclaim.

Watch one worked out

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Work on this practice instance before attempting the transfer questions.

- Record the observation literally.
- State the claim separately.
- Explain exactly how the observation supports the claim.

Does the answer make sense?

- Check that the final conclusion follows from the evidence and rule used.

If you get stuck

Write the observation on one line exactly as it was recorded, write your claim on the line under it, and draw an arrow from one to the other.

Common mistake to avoid

An easy way to get this wrong: A chemically plausible claim is automatically supported.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: A minimal pair differing on exactly one decisive feature.

- Trim unsupported claim content.
- Restate the evidence link.
- Retry on another observation.

Try that one again: Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A

Predict this case before reading on.

CASE B

Predict this case before reading on.

Try it with help

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Use the first-move and picture cues shown here.

Try it with less help

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. One cue has been removed; choose the next move yourself.

Now try it on your own

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Try it independently using only the information in the task.

Check your answer

- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

☐ Check that the final conclusion follows from the evidence and rule used.

Where this shows up next

You will meet this again in: new observation requiring cautious inference. Attempt the transfer questions only after this learning step.

Follow the same substance through a reaction

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.

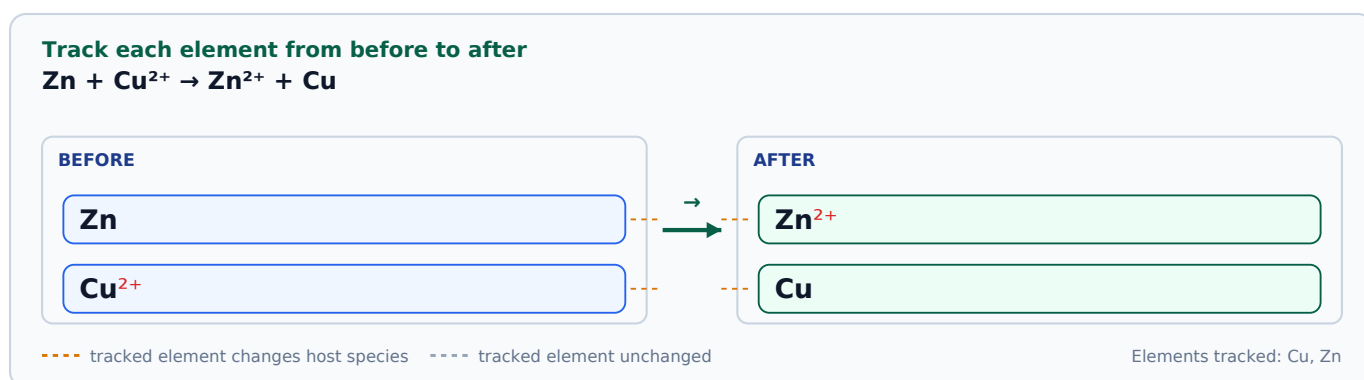
Treat the equation as a before/after map of the same chemical entities.

The idea in plain words

A role is proved from what a species does in this equation, not from reagent reputation.

Picture it like this

- Track changing species first.
- Separate unchanged or spectator information.
- Attach a role only after the change is established.
- Use the symbolic representation.
- Read the formula notation explicitly.
- Read the ionic charge explicitly.



The rule you will use

Equation/source evidence is authoritative; memorized reagent reputation is not.

Why it works

- Identify reactant species.
- Track each through the equation.
- Mark change/no change.
- Attach the role to the reactant evidence.
- Verify spectators are not mislabelled.

Watch one worked out

Track each supplied reacting species from before to after the process before assigning any role label. Work on this practice instance before attempting the transfer questions.

- Identify reactant species.
- Track each corresponding species after the process.
- Mark changed and unchanged species.

Does the answer make sense?

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

If you get stuck

List the starting substances down the left and the finishing ones down the right, then join each substance to itself with a line before you name any role.

Common mistake to avoid

An easy way to get this wrong: A familiar reagent has one permanent role.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: Compare a changed species with an unchanged/spectator species in the same representation.

- Return to species tracking.
- Name the self-change before the role.
- Retry with a new equation/representation.

Try that one again: Track each supplied reacting species from before to after the process before assigning any role label. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A

Zn

Predict this case before reading on.

CASE B

Zn

Predict this case before reading on.

Try it with help

Track each supplied reacting species from before to after the process before assigning any role label. Use the first-move and picture cues shown here.

Try it with less help

Track each supplied reacting species from before to after the process before assigning any role label. One cue has been removed; choose the next move yourself.

Now try it on your own

Track each supplied reacting species from before to after the process before assigning any role label. Try it independently using only the information in the task.

Check your answer

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- ☐ Check that the overall charge is preserved or interpreted correctly.
- ☐ Check that the same chemical species is being tracked.
- ☐ Check that the final conclusion follows from the evidence and rule used.

Where this shows up next

You will meet this again in: same role logic in a new equation. Attempt the transfer questions only after this learning step.

Move between particle pictures and symbols

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.

Begin with the representation the learner can directly inspect before introducing another representation.

The idea in plain words

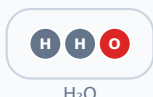
Symbols are a compressed way to describe the same chemical entities; translation must not change what is present.

Picture it like this

- Name the current representation level.
- Hold chemical identity fixed while translating.
- Return to the start representation to check equivalence.
- Use the particulate representation.
- Use the symbolic representation.
- Read the figure as part of the evidence.
- Read the formula notation explicitly.

Particle view of the same substance

School-level particle-count model — composition and count only.



● H × 2 ● O × 1

1 particle(s) shown

Same chemical entity across three views

Which chemical entity remains the same across representation levels.

Stated / observed

Connect macroscopic wording or observation to particle and symbolic representations while preserving identity.

Particle model



identity and count preserved

Symbolic notation



same identity, written down

Trace the same entity through macro, particle and symbolic panels.

The rule you will use

Preserve identity, count and any recorded condition while changing representation.

Why it works

- Identify the source representation.
- Mark invariant entities.
- Translate one feature at a time.
- Cross-check against the starting representation.

Watch one worked out

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Work on this practice instance before attempting the transfer questions.

- Identify the starting representation level.
- Mark invariant entities/counts.
- Translate one feature at a time.
- Cross-check against the starting representation.

Does the answer make sense?

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

If you get stuck

Count the separate particles you can actually see and write that number down before you write any symbols.

Common mistake to avoid

An easy way to get this wrong: Changing notation is harmless if the picture looks similar.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: Compare a faithful translation with one that changes a subscript, charge or species identity.

- Return to the source representation.
- Repair the first changed feature.
- Retry in the reverse direction.

Try that one again: Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A



Predict this case before reading on.

CASE B



Predict this case before reading on.

Try it with help

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Use the first-move and picture cues shown here.

Try it with less help

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. One cue has been removed; choose the next move yourself.

Now try it on your own

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Try it independently using only the information in the task.

Check your answer

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- ☐ Check that the same chemical species is being tracked.
- ☐ Check that the final conclusion follows from the evidence and rule used.

Where this shows up next

You will meet this again in: same chemistry in a different representation. Attempt the transfer questions only after this learning step.

Check your chemistry before you accept an answer

Get ready

Identify the target and state the first chemically meaningful move: Identify the target.

Make checking a separate pass, not a repetition of the final answer.

The idea in plain words

A tidy final answer is not chemically verified until an independent check is applied.

Picture it like this

- Run only applicable checks: species identity, atoms, charge, units, state/condition and plausibility.
- Use the symbolic representation.

Run these checks before accepting

- ☐ Check that the same chemical species is being tracked.
- ☐ Check that the final conclusion follows from the evidence and rule used.

The rule you will use

Which checks you must run comes from the question itself and from what this page has taught you.

Why it works

- Name the required checks.
- Run each independently.
- Locate any failure.
- Repair.
- Re-run.

Watch one worked out

Run the required independent chemical checks on the supplied result or representation before accepting it. Work on this practice instance before attempting the transfer questions.

- Name the required verification checks.
- Run each check independently.
- Repair any failed check and re-run it.

Does the answer make sense?

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

If you get stuck

List the checks that apply as short lines, then tick them off one at a time against your own working.

Common mistake to avoid

An easy way to get this wrong: Getting an answer means it has been checked.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Now compare these two: Contrast a neat-looking result with one that passes the required chemical invariant.

- Choose one independent check.

- Apply it.
- Repair and repeat.

Try that one again: Run the required independent chemical checks on the supplied result or representation before accepting it. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A

Predict this case before reading on.

CASE B

Predict this case before reading on.

Try it with help

Run the required independent chemical checks on the supplied result or representation before accepting it. Use the first-move and picture cues shown here.

Try it with less help

Run the required independent chemical checks on the supplied result or representation before accepting it. One cue has been removed; choose the next move yourself.

Now try it on your own

Run the required independent chemical checks on the supplied result or representation before accepting it. Try it independently using only the information in the task.

Check your answer

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

☐ Check that the same chemical species is being tracked.

☐ Check that the final conclusion follows from the evidence and rule used.

Where this shows up next

You will meet this again in: checking your work on a new kind of question. Attempt the transfer questions only after this learning step.

Practice questions

Practice 1 — Name the role only after you have tracked the change

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Use the first-move and picture cues shown here.

Your working space: _____

Practice 2 — Name the role only after you have tracked the change

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 3 — Name the role only after you have tracked the change

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Try it independently using only the information in the task.

Your working space: _____

Practice 4 — Check that the atoms balance

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Use the first-move and picture cues shown here.

Your working space: _____

Practice 5 — Check that the atoms balance

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 6 — Check that the atoms balance

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Try it independently using only the information in the task.

Your working space: _____

Practice 7 — Check the rule, the condition and the exception

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Use the first-move and picture cues shown here.

Your working space: _____

Practice 8 — Check the rule, the condition and the exception

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 9 — Check the rule, the condition and the exception

Apply the supplied rule only after checking the recorded condition or exception.

State which feature controls the decision. Try it independently using only the information in the task.

Your working space: _____

Practice 10 — Decide what kind of change happened, from the evidence

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Use the first-move and picture cues shown here.

Your working space: _____

Practice 11 — Decide what kind of change happened, from the evidence

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 12 — Decide what kind of change happened, from the evidence

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Try it independently using only the information in the task.

Your working space: _____

Practice 13 — Tell an ion charge apart from a subscript

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Use the first-move and picture cues shown here.

Your working space: _____

Practice 14 — Tell an ion charge apart from a subscript

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 15 — Tell an ion charge apart from a subscript

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Try it independently using only the information in the task.

Your working space: _____

Practice 16 — Keep the reaction conditions attached

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Use the first-move and picture cues shown here.

Your working space: _____

Practice 17 — Keep the reaction conditions attached

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 18 — Keep the reaction conditions attached

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Try it independently using only the information in the task.

Your working space: _____

Practice 19 — Connect the apparatus, the property and the method

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Use the first-move and picture cues shown here.

Your working space: _____

Practice 20 — Connect the apparatus, the property and the method

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 21 — Connect the apparatus, the property and the method

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Try it independently using only the information in the task.

Your working space: _____

Practice 22 — Read a chemical formula before you calculate

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Use the first-move and picture cues shown here.

Your working space: _____

Practice 23 — Read a chemical formula before you calculate

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 24 — Read a chemical formula before you calculate

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Try it independently using only the information in the task.

Your working space: _____

Practice 25 — Keep what you saw apart from what it means

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Use the first-move and picture cues shown here.

Your working space: _____

Practice 26 — Keep what you saw apart from what it means

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 27 — Keep what you saw apart from what it means

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Try it independently using only the information in the task.

Your working space: _____

Practice 28 — Follow the same substance through a reaction

Track each supplied reacting species from before to after the process before assigning any role label. Use the first-move and picture cues shown here.

Your working space: _____

Practice 29 — Follow the same substance through a reaction

Track each supplied reacting species from before to after the process before assigning any role label. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 30 — Follow the same substance through a reaction

Track each supplied reacting species from before to after the process before assigning any role label. Try it independently using only the information in the task.

Your working space: _____

Practice 31 — Move between particle pictures and symbols

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Use the first-move and picture cues shown here.

Your working space: _____

Practice 32 — Move between particle pictures and symbols

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. One cue has been removed; choose the next move yourself.

Your working space: _____

Practice 33 — Move between particle pictures and symbols

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Try it independently using only the information in the task.

Your working space: _____

Practice 34 — Check your chemistry before you accept an answer

Run the required independent chemical checks on the supplied result or representation before accepting it. Use the first-move and picture cues shown here.

Your working space: _____

Practice 35 — Check your chemistry before you accept an answer

Run the required independent chemical checks on the supplied result or representation before accepting it. One cue has been removed; choose the next

move yourself.

Your working space: _____

Practice 36 — Check your chemistry before you accept an answer

Run the required independent chemical checks on the supplied result or representation before accepting it. Try it independently using only the information in the task.

Your working space: _____

Quick answers — cover this page while you work

Practice 1

Check your working shows all 4 points: track the reactant species first; use its recorded change/effect as evidence; attach the requested role and exclude spectators; runs every required check listed with the worked answer.

Practice 2

Check your working shows all 4 points: track the reactant species first; use its recorded change/effect as evidence; attach the requested role and exclude spectators; runs every required check listed with the worked answer.

Practice 3

Check your working shows all 4 points: track the reactant species first; use its recorded change/effect as evidence; attach the requested role and exclude spectators; runs every required check listed with the worked answer.

Practice 4

Check your working shows all 4 points: choose the atom-count ledger; count each required atom before and after; compare counts and identify any mismatch; runs every required check listed with the worked answer.

Practice 5

Check your working shows all 4 points: choose the atom-count ledger; count each required atom before and after; compare counts and identify any mismatch; runs every required check listed with the worked answer.

Practice 6

Check your working shows all 4 points: choose the atom-count ledger; count each required atom before and after; compare counts and identify any mismatch; runs every required check listed with the worked answer.

Practice 7

Check your working shows all 4 points: select the recorded rule; check the condition/exception gate; apply or withhold the rule accordingly; runs every required check listed with the worked answer.

Practice 8

Check your working shows all 4 points: select the recorded rule; check the condition/exception gate; apply or withhold the rule accordingly; runs every required check listed with the worked answer.

Practice 9

Check your working shows all 4 points: select the recorded rule; check the condition/exception gate; apply or withhold the rule accordingly; runs every required check listed with the worked answer.

Practice 10

Check your working shows all 4 points: extract the supplied evidence; apply the authorized criterion; classify only after the evidence matches the criterion; runs every required check listed with the worked answer.

Practice 11

Check your working shows all 4 points: extract the supplied evidence; apply the authorized criterion; classify only after the evidence matches the criterion; runs every required check listed with the worked answer.

Practice 12

Check your working shows all 4 points: extract the supplied evidence; apply the authorized

criterion; classify only after the evidence matches the criterion; runs every required check listed with the worked answer.

Practice 13

Check your working shows all 4 points: locate the superscript charge separately from subscripts/coefficients; read the ionic charge; verify the species identity is unchanged; runs every required check listed with the worked answer.

Practice 14

Check your working shows all 4 points: locate the superscript charge separately from subscripts/coefficients; read the ionic charge; verify the species identity is unchanged; runs every required check listed with the worked answer.

Practice 15

Check your working shows all 4 points: locate the superscript charge separately from subscripts/coefficients; read the ionic charge; verify the species identity is unchanged; runs every required check listed with the worked answer.

Practice 16

Check your working shows all 4 points: locate the recorded condition; keep it attached to the process representation; check that the rewritten/read representation preserves it; runs every required check listed with the worked answer.

Practice 17

Check your working shows all 4 points: locate the recorded condition; keep it attached to the process representation; check that the rewritten/read representation preserves it; runs every required check listed with the worked answer.

Practice 18

Check your working shows all 4 points: locate the recorded condition; keep it attached to the process representation; check that the rewritten/read representation preserves it; runs every required check listed with the worked answer.

Practice 19

Check your working shows all 4 points: identify the target property or observation; identify the apparatus/method feature; explain the property-feature link; runs every required check listed with the worked answer.

Practice 20

Check your working shows all 4 points: identify the target property or observation; identify the apparatus/method feature; explain the property-feature link; runs every required check listed with the worked answer.

Practice 21

Check your working shows all 4 points: identify the target property or observation; identify the apparatus/method feature; explain the property-feature link; runs every required check listed with the worked answer.

Practice 22

Check your working shows all 4 points: locate each notation feature by position; state the local meaning of each feature; check the complete species notation; runs every required check listed with the worked answer.

Practice 23

Check your working shows all 4 points: locate each notation feature by position; state the local meaning of each feature; check the complete species notation; runs every required check listed with the worked answer.

Practice 24

Check your working shows all 4 points: locate each notation feature by position; state the local

meaning of each feature; check the complete species notation; runs every required check listed with the worked answer.

Practice 25

Check your working shows all 4 points: record the observation literally; state the claim separately; explain exactly how the observation supports the claim; runs every required check listed with the worked answer.

Practice 26

Check your working shows all 4 points: record the observation literally; state the claim separately; explain exactly how the observation supports the claim; runs every required check listed with the worked answer.

Practice 27

Check your working shows all 4 points: record the observation literally; state the claim separately; explain exactly how the observation supports the claim; runs every required check listed with the worked answer.

Practice 28

Check your working shows all 4 points: identify reactant species; track each corresponding species after the process; mark changed and unchanged species; runs every required check listed with the worked answer.

Practice 29

Check your working shows all 4 points: identify reactant species; track each corresponding species after the process; mark changed and unchanged species; runs every required check listed with the worked answer.

Practice 30

Check your working shows all 4 points: identify reactant species; track each corresponding species after the process; mark changed and unchanged species; runs every required check listed with the worked answer.

Practice 31

Check your working shows all 5 points: identify the starting representation level; mark invariant entities/counts; translate one feature at a time; cross-check against the starting representation; runs every required check listed with the worked answer.

Practice 32

Check your working shows all 5 points: identify the starting representation level; mark invariant entities/counts; translate one feature at a time; cross-check against the starting representation; runs every required check listed with the worked answer.

Practice 33

Check your working shows all 5 points: identify the starting representation level; mark invariant entities/counts; translate one feature at a time; cross-check against the starting representation; runs every required check listed with the worked answer.

Practice 34

Check your working shows all 4 points: name the required verification checks; run each check independently; repair any failed check and re-run it; runs every required check listed with the worked answer.

Practice 35

Check your working shows all 4 points: name the required verification checks; run each check independently; repair any failed check and re-run it; runs every required check listed with the worked answer.

Practice 36

Check your working shows all 4 points: name the required verification checks; run each check

independently; repair any failed check and re-run it; runs every required check listed with the worked answer.

Worked answers

Worked answer 1 — Name the role only after you have tracked the change

- Track the reactant species first.
- Use its recorded change/effect as evidence.
- Attach the requested role and exclude spectators.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 2 — Name the role only after you have tracked the change

- Track the reactant species first.
- Use its recorded change/effect as evidence.
- Attach the requested role and exclude spectators.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 3 — Name the role only after you have tracked the change

- Track the reactant species first.
- Use its recorded change/effect as evidence.
- Attach the requested role and exclude spectators.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 4 — Check that the atoms balance

- Choose the atom-count ledger.
- Count each required atom before and after.
- Compare counts and identify any mismatch.
- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 5 — Check that the atoms balance

- Choose the atom-count ledger.
- Count each required atom before and after.

- Compare counts and identify any mismatch.
- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 6 — Check that the atoms balance

- Choose the atom-count ledger.
- Count each required atom before and after.
- Compare counts and identify any mismatch.
- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 7 — Check the rule, the condition and the exception

- Select the recorded rule.
- Check the condition/exception gate.
- Apply or withhold the rule accordingly.
- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve the explicit exception supplied with the task.

Worked answer 8 — Check the rule, the condition and the exception

- Select the recorded rule.
- Check the condition/exception gate.
- Apply or withhold the rule accordingly.
- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve the explicit exception supplied with the task.

Worked answer 9 — Check the rule, the condition and the exception

- Select the recorded rule.
- Check the condition/exception gate.
- Apply or withhold the rule accordingly.
- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve the explicit exception supplied with the task.

Worked answer 10 — Decide what kind of change happened, from the evidence

- Extract the supplied evidence.

- Apply the authorized criterion.
- Classify only after the evidence matches the criterion.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 11 — Decide what kind of change happened, from the evidence

- Extract the supplied evidence.
- Apply the authorized criterion.
- Classify only after the evidence matches the criterion.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 12 — Decide what kind of change happened, from the evidence

- Extract the supplied evidence.
- Apply the authorized criterion.
- Classify only after the evidence matches the criterion.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 13 — Tell an ion charge apart from a subscript

- Locate the superscript charge separately from subscripts/coefficients.
- Read the ionic charge.
- Verify the species identity is unchanged.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 14 — Tell an ion charge apart from a subscript

- Locate the superscript charge separately from subscripts/coefficients.
- Read the ionic charge.
- Verify the species identity is unchanged.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 15 — Tell an ion charge apart from a subscript

- Locate the superscript charge separately from subscripts/coefficients.
- Read the ionic charge.
- Verify the species identity is unchanged.

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 16 — Keep the reaction conditions attached

- Locate the recorded condition.
- Keep it attached to the process representation.
- Check that the rewritten/read representation preserves it.
- Check that every stated condition or exception is still attached.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve and apply: heat

Worked answer 17 — Keep the reaction conditions attached

- Locate the recorded condition.
- Keep it attached to the process representation.
- Check that the rewritten/read representation preserves it.
- Check that every stated condition or exception is still attached.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve and apply: heat

Worked answer 18 — Keep the reaction conditions attached

- Locate the recorded condition.
- Keep it attached to the process representation.
- Check that the rewritten/read representation preserves it.
- Check that every stated condition or exception is still attached.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve and apply: heat

Worked answer 19 — Connect the apparatus, the property and the method

- Identify the target property or observation.
- Identify the apparatus/method feature.
- Explain the property-feature link.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 20 — Connect the apparatus, the property and the method

- Identify the target property or observation.
- Identify the apparatus/method feature.
- Explain the property-feature link.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 21 — Connect the apparatus, the property and the method

- Identify the target property or observation.
- Identify the apparatus/method feature.
- Explain the property-feature link.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 22 — Read a chemical formula before you calculate

- Locate each notation feature by position.
- State the local meaning of each feature.
- Check the complete species notation.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 23 — Read a chemical formula before you calculate

- Locate each notation feature by position.
- State the local meaning of each feature.
- Check the complete species notation.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 24 — Read a chemical formula before you calculate

- Locate each notation feature by position.
- State the local meaning of each feature.
- Check the complete species notation.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 25 — Keep what you saw apart from what it means

- Record the observation literally.
- State the claim separately.
- Explain exactly how the observation supports the claim.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 26 — Keep what you saw apart from what it means

- Record the observation literally.
- State the claim separately.
- Explain exactly how the observation supports the claim.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 27 — Keep what you saw apart from what it means

- Record the observation literally.
- State the claim separately.
- Explain exactly how the observation supports the claim.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 28 — Follow the same substance through a reaction

- Identify reactant species.
- Track each corresponding species after the process.
- Mark changed and unchanged species.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 29 — Follow the same substance through a reaction

- Identify reactant species.
- Track each corresponding species after the process.
- Mark changed and unchanged species.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 30 — Follow the same substance through a reaction

- Identify reactant species.
- Track each corresponding species after the process.
- Mark changed and unchanged species.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded

reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 31 — Move between particle pictures and symbols

- Identify the starting representation level.
- Mark invariant entities/counts.
- Translate one feature at a time.
- Cross-check against the starting representation.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 32 — Move between particle pictures and symbols

- Identify the starting representation level.
- Mark invariant entities/counts.
- Translate one feature at a time.
- Cross-check against the starting representation.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 33 — Move between particle pictures and symbols

- Identify the starting representation level.
- Mark invariant entities/counts.
- Translate one feature at a time.
- Cross-check against the starting representation.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 34 — Check your chemistry before you accept an answer

- Name the required verification checks.
- Run each check independently.
- Repair any failed check and re-run it.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 35 — Check your chemistry before you accept an answer

- Name the required verification checks.
- Run each check independently.

- Repair any failed check and re-run it.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

Worked answer 36 — Check your chemistry before you accept an answer

- Name the required verification checks.
- Run each check independently.
- Repair any failed check and re-run it.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No extra condition or exception applies to this task.

One-page reminder sheet

Print this page. It has no answers on it, and nothing new to learn — it is just the reminders.

Name the role only after you have tracked the change

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that the overall charge is preserved or interpreted correctly.

Check that the atoms balance

Start by: Identify the target.

Rule to remember: Choose the conservation check the question actually needs.

Before you move on: Count each required atom on both sides.

Check the rule, the condition and the exception

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that every stated condition or exception is still attached.

Decide what kind of change happened, from the evidence

Start by: Identify the target.

Rule to remember: Classify only from the stated evidence and the rule taught here.

Before you move on: Check that the final conclusion follows from the evidence and rule used.

Tell an ion charge apart from a subscript

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that the overall charge is preserved or interpreted correctly.

Keep the reaction conditions attached

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that every stated condition or exception is still attached.

Connect the apparatus, the property and the method

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that the final conclusion follows from the evidence and rule used.

Read a chemical formula before you calculate

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that the overall charge is preserved or interpreted correctly.

Keep what you saw apart from what it means

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that the final conclusion follows from the evidence and rule used.

Follow the same substance through a reaction

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that the overall charge is preserved or interpreted correctly.

Move between particle pictures and symbols

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that the same chemical species is being tracked.

Check your chemistry before you accept an answer

Start by: Identify the target.

Rule to remember: Use only the chemistry stated or taught for this task.

Before you move on: Check that the same chemical species is being tracked.